

Dishevelled2 Is a Stable Protein During Early Zebrafish Development

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Abstract

Wnt signaling is a major player during development and its misregulation often leads to disease, especially cancer. The negative feedback Wnt regulator homologs, Nkd1 and Nkd2, have been shown to inhibit Wnt signaling during development, and current evidence suggests that Nkds degrade Dvl proteins to antagonize Wnt signaling. Here, we demonstrate that during early zebrafish development Nkd1 does not alter either endogenous or exogenous levels of Dvl2. Furthermore, Dvl2 does not affect the levels of Nkd1. Cumulatively, these results demonstrate that Dvl2 is a ubiquitous and stable protein and that Nkds may not always function to degrade Dvl proteins as a method of inhibiting Wnt signaling.

Introduction

DROSOPHILA NAKED CUTICLE (Nkd) and its vertebrate homologs Naked1 (Nkd1) and Naked2 (Nkd2) antagonize Wnt signaling, and recent *in vitro* evidence suggests that Nkds degrade Dvl, providing an effective way to shut down the pathway.^{1–3} However, there are functional differences between Nkd proteins to suggest that they are not completely redundant.^{4,5} For example, Nkd2 has an additional function in epidermal growth factor receptor (EGFR) signaling.^{6,7} In contrast, Nkd1 appears specific for Wnt signaling and is more reminiscent of *Drosophila* Nkd, both being Wnt-induced negative regulators.^{5,8} Currently, there is no evidence that fly Nkd degrades Dvl, and there is new evidence to suggest that Nkd1 antagonizes Wnt signaling independent of Dvl.^{4,9}

As Nkd and Nkd1 have been shown to bind other Dvl proteins,^{4,8,10} we wanted to further investigate the role of Nkd1 in the stability of Dvl2. To investigate the interaction between Nkd1 and Dvl2, we used the early zebrafish embryo for which the Wnt signaling pathways and timing of Wnt signaling are well characterized.^{11–13} To investigate endogenous Dvl2, a polyclonal antibody to zebrafish Dvl2 was generated. In contrast to recent findings,^{1–3} this study demonstrates that during early zebrafish development Dvl2 is a stable protein and is not involved in hastening the degradation of Nkd1. Nkd1 also does not increase the degradation rate of either endogenous or exogenous Dvl2.

Results

Characterization of anti-Dvl2 antibody

To investigate the role of endogenous Dvl2, a polyclonal antibody to zebrafish Dvl2 was generated. The epitope was

designed in a nonconserved region of Dvl2 and is not predicted to cross react with Dvl1 or Dvl3 (Fig. 1A), although this has not been formally tested. Affinity-purified polyclonal anti-Dvl2 can detect both exogenous and endogenous Dvl2 proteins when used as low as 1:20,000 for western analysis (Fig. 1B). Knockdown of endogenous Dvl2 with translation blocking antisense oligonucleotide morpholinos (MO) was observed with the antibody, confirming that the antibody can detect endogenous Dvl2 (Fig. 1C, D). To determine if the antibody could detect exogenous Dvl2 by immunohistochemistry, Dvl2^{HA} was injected into one of four blastomeres to generate mosaic Dvl2^{HA} expression. The antibody could detect Dvl2^{HA} at a dilution of 1:5000 with similar, but weaker staining in non-Dvl2^{HA}-expressing cells (Fig. 1F, I). The HA and the Dvl2 antibodies give nearly identical staining patterns and detect cellular puncta, which is characteristic of Dvl staining and is very similar to *Drosophila* Dsh and mammalian Dvl2 proteins.^{14–17} At a higher concentration (1:1000), endogenous Dvl2 staining is very similar to that seen at 1:5000 in both Dvl2^{HA}-expressing and -nonexpressing cells; only the staining is more robust and is ubiquitous (Fig. 1G). In a co-labeling experiment, there is substantial overlap between the HA and the Dvl2 antibodies (Fig. 1H–J). We conclude that the antibody to zebrafish Dvl2 is robust and can detect endogenous Dvl2 by both western analysis and immunohistochemistry.

Dvl2 is stable, but Nkd1 is not

Previously, it has been demonstrated in cell culture that Dvl proteins are targeted for degradation via several mechanisms.^{1,3,10,16,18,19} Therefore, we wanted to determine the